

Questions by topic: Manipulation of food species & agricultural energetics - Set 1 (65 marks)

Answer all questions in the spaces provided.

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Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0 1 . 2

Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

10

0 2 Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 2 . **1** Explain the main advantage of GM over conventional selective breeding.

[2 marks]

0 2 . **2** For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops

Roundup Ready soya

Golden rice

0 2 . **3** Explain **four** concerns about GM crops.

[4 marks]

0 2 . 4 Give **one** reason why GM crops have been slow to enter cultivation in Europe. [1 mark]

10

0 3 Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha ⁻¹ yr ⁻¹	Food energy output / GJ ha ⁻¹ yr ⁻¹
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 3 . 1 Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 3 . 2 Use your answers to explain the difference between productivity and efficiency.

[3 marks]

0 3 . 3

Explain what is meant by an **energy subsidy** and give **three** examples.**[3 marks]**

0 3 . 4

Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]**10**

0 4

Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 4 . 1

Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

0 4 . 2

Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

0 4 . 3

Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

10

0 5 . **1** State what is meant by **hybrid vigour**.

[1 mark]

0 5 . **2** Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

0 5 . **3** Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

2

5

0 6 Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 6 . **1** Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer

%

06 . 2

Explain why so little of the solar energy is stored in the crop.

[3 marks]

06 . 3

Explain why few food chains have more than four trophic levels.

[2 marks]

06 . 4

Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

06 . 5

Explain how pigs can increase the efficiency of a food production system.

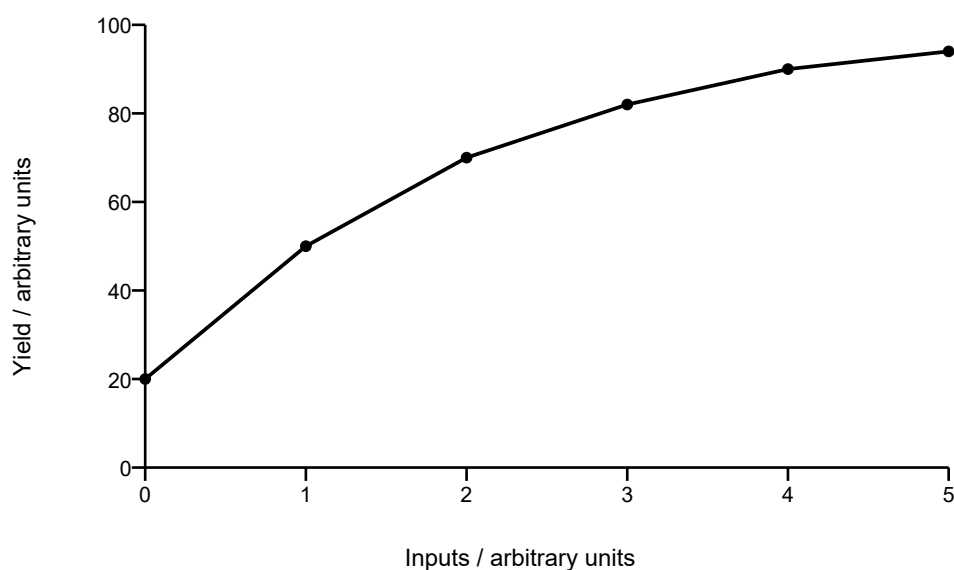
[2 marks]

10

07

Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



07

. 1 Describe the relationship shown in Figure 1.

[2 marks]

07

. 2 Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer

units

07 . 3

Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

07 . 4

Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

07 . 5

Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

10

END OF QUESTIONS